

《计算机网络》作业

第二次作业：（英文版教材第二章 2, 3, 4, 7, 8, 教材第四章 14)

2(2.1.3 The Maximum Data Rate of a Channel). A noiseless 8-kHz channel is sampled every 1 msec. What is the maximum data rate?

3(2.1.3 The Maximum Data Rate of a Channel). If a binary signal is sent over a 3-kHz channel whose signal-to-noise ratio is 20 dB, what is the maximum achievable data rate?

4(2.1.3 The Maximum Data Rate of a Channel). What signal-to-noise ratio is needed to put a T1 carrier with data rate 1.544Mbps on a 100-kHz line?

7(2.6.4 Trunks and Multiplexing-Digitizing Analog Signals). It is desired to send a sequence of computer screen images over an optical fiber. The screen is 1920×1200 pixels, each pixel being 24 bits. There are 50 screen images per second. How much bandwidth is needed?

8(2.1.3 The Maximum Data Rate of a Channel). Is the Nyquist theorem true for high-quality single-mode optical fiber or only for copper wire?

4-14(2.5.1 Baseband Transmission) Sketch the Manchester encoding on a classic Ethernet for the bit stream 0001110101.